JEFFRIN SINGH R

AI Engineer | Computer Vision | Backend Systems

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building real-time computer vision pipelines and production-grade backend systems. Skilled in YOLO, InsightFace, ONNX Runtime, and FastAPI. Achieved approximately 30% latency reduction and 20 FPS on live 720p streams in real-world deployments.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI and Computer Vision: YOLO, InsightFace, OpenCV, PaddleOCR, EasyOCR, ONNX Runtime, ByteTrack

Backend: FastAPI, Flask, REST APIs, WebSockets

Frontend: React, Vite, Tailwind CSS, HTML, CSS

Databases: PostgreSQL, MySQL, SQL Server, Supabase

Tools and DevOps: Docker, Git, Vercel, Render

EXPERIENCE

AI Engineer Intern, Invendoi AI Solutions | Feb 2026 - Apr 2026

Kochi, India

- Built a real-time Automatic Number Plate Recognition (ANPR) system using YOLO, ByteTrack, and PaddleOCR on live video streams, enabling automated vehicle tracking with minimized duplicate detections across multi-camera setups.
- Optimised inference pipeline with ONNX Runtime, reducing per-frame latency by approximately 30% and achieving approximately 20 FPS on 720p streams, enabling real-time deployment in production surveillance environments.
- Developed and deployed three production-grade platforms used by real users: Invendoi (React, Vercel, Supabase), Invendo AI Academy (React, Vercel, Render, Supabase, Gemini API-powered AI chatbot), and Campus Platform (HTML, CSS, JavaScript), optimized for performance and responsiveness.

PROJECTS

AI Document Intelligence System | 2026

Tech Stack: PaddleOCR (PP-OCRv4), PyMuPDF, Vision-Language Models, Tkinter, Python

- Built an end-to-end OCR pipeline for images and multi-page PDFs using PaddleOCR (PP-OCRv4 detector and recognizer) with automatic GPU/CPU fallback, reading-order reconstruction, and text normalization, exporting structured JSON, normalized JSON, and plain-text outputs.
- Extended it with a vision-language grounding module (nvidia/LocateAnything-3B) to locate elements such as logos, signatures, stamps, and tables via free-text queries, plus a Tkinter GUI orchestrating both pipelines with live progress and performance logging.

AI Face Recognition System | 2026

Tech Stack: InsightFace, FastAPI, PostgreSQL, Docker, Python

- Designed and built a real-time face recognition pipeline using InsightFace with cosine similarity matching, achieving sub-100ms response latency per query.
- Developed a FastAPI backend with PostgreSQL to log all recognition events, enabling audit trails and analytics; containerized the system with Docker for multi-camera scalability.

Automatic Number Plate Recognition (ANPR) System | 2026

Tech Stack: YOLO, ByteTrack, PaddleOCR, Python, OpenCV

- Built an end-to-end ANPR pipeline integrating YOLO for plate detection, ByteTrack for consistent multi-frame vehicle tracking, and PaddleOCR for text extraction under real-world lighting conditions.
- Applied preprocessing techniques such as contrast normalisation and denoising to improve OCR accuracy on low-quality frames, making the system production-ready for surveillance use cases.

Secure Payment and Subscription System | 2024

Tech Stack: Flask, Razorpay API, PostgreSQL, Python

- Built a Flask web application with Razorpay payment gateway integration, implementing full authentication, subscription validation, and payment workflow to support end-to-end user onboarding.
- Designed a normalised PostgreSQL schema for user and payment management, ensuring data integrity and enabling subscription lifecycle tracking.

EDUCATION

B.E., Computer Science and Engineering, Loyola Institute of Technology and Science

Graduating 2026 | *Tamil Nadu, India*

CERTIFICATIONS

AI/ML Training Program, Feather Softwares | Web Development Internship, Roriri Software Solutions | Python for Data Science, Udemy | Bootstrap Mastery, Udemy